Quadratic Forms and Optimization*

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1 Quadratic Forms

It is important to approximate nonlinear functions not only with linear (affine) functions, but also with quadratic functions. For example, for a real function $f : \mathbb{R} \rightarrow \mathbb{R}$, our quadratic approximating function will be $f(\bar{x})+f'(\bar{x})\Delta x+\frac{1}{2}f"(x)(\Delta x)^2$. The quadratic approximation is therefore the quadratic function that has (i) the same value as f at $\bar{x}$, (ii) the same slope (the same derivative) as f at $\bar{x}$, and (iii) the same curvature (the same second derivative) as f at $\bar{x}$.

When we generalize from functions with the one-dimensional domain $\mathbb{R}$ to multivariate functions, with domain $\mathbb{R}^n$, it becomes more complicated. The derivative of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at a point $\bar{\mathbf{x}} \in \mathbb{R}^n$ is no longer just a number, but a vector - specifically, the gradient of f at $\bar{\mathbf{x}}$, which we write as $\nabla f(\bar{\mathbf{x}})$. And the quadratic term in the quadratic approximation to f is a *quadratic form*, which is defined by an $n \times n$ matrix $H(\bar{\mathbf{x}}$ - the second derivative of f at $\bar{\mathbf{x}}$.

Quadratic function is a second degree polynomial function, $f(x) = ax^2 + bx + c$ in which $a \neq 0$. If each of the coefficients, a, b and c is non-zero, then the function has a second-degree (quadratic) term, a first-degree (linear) term, and a zero-degree (constant) term. However, a *quadratic form* is a real-valued function on $\mathbb{R}^n$ that has *only* second degree (quadratic) terms. So a quadratic function on $\mathbb{R}$ (i.e., on $\mathbb{R}^n$, where $n = 1$) is a function of the form $f(x) = ax^2$ for some non-zero coefficient $a \in \mathbb{R}$.

When $n = 2$, a quadratic form on $\mathbb{R}^2$ is a function of the form $f(x_1, x_2) = a_{11}x_1 + a_{12}x_1x_2 + a_{21}x_2x_1 + a_{22}x_2x_2$, or equivalently, $f(x_1, x_2) = a_{11}x_1^2 + (a_{12} + a_{21})x_1x_2 + a_{22}x_2^2$.

*This material is largely based on the lecture notes of math camp in Arizona University and Columbia University.

Note that in the second expression for f we combined the coefficients a_{12} and a_{21} into their sum, so we can also write the same function as $f(x_1, x_2) = a_{11}x_1^2 + a'_{12}x_1x_2 + a_{22}x_2^2$, where $a'_{12} = a_{12} + a_{21}$, which is a common way to write quadratic forms.

The quadratic form on $\mathbb{R}^2$ can also be written as

$$f(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Moreover, without loss of generality, we can assume that $a_{12} = a_{21}$ (if a_{12} and a_{21} are not equal, we can write them instead as $\tilde{a}_{12}$ and $\tilde{a}_{21}$ and then define $a_{12} = a_{21} = \frac{\tilde{a}_{21}+\tilde{a}_{12}}{2}$. Therefore, the quadratic forms on $\mathbb{R}^2$ are precisely the function $f : \mathbb{R}^2 \to \mathbb{R}$ of the form

$$f(\mathbf{x}) = f(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{x}^T A\mathbf{x} \tag{1}$$

where A is a symmetric matrix.

Before moving to the general case of $\mathbb{R}^n$, let's consider the case of $\mathbb{R}^3$. In this case, the generic quadratic form is

$$\begin{aligned} f(x_1, x_2, x_3) = a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + a_{12}x_1x_2 + a_{21}x_2x_1 \\ + a_{13}x_1x_3 + a_{31}x_3x_1 + a_{23}x_2x_3 + a_{32}x_3x_2, \end{aligned}$$

and we can assume, as before, that $a_{12} = a_{21}$, $a_{13} = a_{31}$, and $a_{23} = a_{32}$. Therefore, we can write the quadratic form as

$$f(\mathbf{x}) = f(x_1, x_2, x_3) = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{x}^T A\mathbf{x}$$

where A is a symmetric 3×3 matrix.

Definition 1.1. *A quadratic form on $\mathbb{R}^n$ is a function $f : \mathbb{R}^n \to \mathbb{R}$ of the form $f(\boldsymbol{x}) = \boldsymbol{x}^T A\boldsymbol{x}$, where A is a symmetric $n \times n$ matrix.*

Since the quadratic form corresponds to a unique symmetric matrix, we can characterize various classes of quadratic forms completely in terms of properties of symmetric matrices. By identifying the matrix A, we can answer some questions as: which quadratic forms on $\mathbb{R}^n$ always have nonnegative values, or which ones are strictly concave functions, etc. In the simple case, $n = 1$, i.e., $f(x) = ax^2$, if $a > 0$ this quadratic form is positive for all nonzero values of x and if $a < 0$ the quadratic form is negative for all nonzero values of x. Moreover, in the $a > 0$ case, the function $f(\cdot)$ is strictly convex and when $a < 0$, the function is strictly concave.

Coming back to $n = 2$, as in Eq. (1), we want to know about whether it is negative (or at least non-positive) for all $\mathbf{x} \neq \mathbf{0}$; or whether neither of these is true, i.e., it is positive for some vectors and negative for others. If we say that it is positive (negative) for all nonzero vectors, we say that the quadratic form is **positive definite** (**negative definite**); if all we can say is that it is non-negative (non-positive) for all nonzero vectors, we say the quadratic form is **positive semi-definite** (**negative semi-definite**). If the sign can go either way, we say the quadratic form is **indefinite**.

Exercise 1 (SB 13.11)**.** *Write the following linear functions in matrix form:*

1. $f(x_1, x_2, x_3) = 2x_1 - 3x_2 + 5x_3$;
2. $f(x_1, x_2) = (2x_1 - 3x_2, x_1 - 4x_2, x_1)$;
3. $f(x_1, x_2, x_3) = (x_1 - x_3, 2x_1 + 3x_2 - 6x_3, x_3 + 2x_2)$;

Exercise 2 (SB 13.12)**.** *Write the following quadratic forms in matrix form:*

1. $x_1^2 - 2x_1x_2 + x_2^2$,
2. $5x_1^2 - 10x_1x_2 - x_2^2$,
3. $x_1^2 + 2x_2^2 + 3x_3^2 + 4x_1x_2 - 6x_1x_3 + 8x_2x_3$.

Definition 1.2. *Let A be an $n \times n$ symmetric matrix, then A is:*

(a) ***positive definite*** *if $\boldsymbol{x}^T A\boldsymbol{x} > 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$ in $\mathbb{R}^n$.*

(b) ***positive semidefinite*** *if $\boldsymbol{x}^T A\boldsymbol{x} \geq 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$ in $\mathbb{R}^n$.*

(c) ***negative definite*** *if $\boldsymbol{x}^T A\boldsymbol{x} < 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$ in $\mathbb{R}^n$.*

(d) ***negative semidefinite*** *if $\boldsymbol{x}^T A\boldsymbol{x} \leq 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$ in $\mathbb{R}^n$.*

(e) ***indefinite*** *if $\boldsymbol{x}^T A\boldsymbol{x} > 0$ for some $\boldsymbol{x}$ in $\mathbb{R}^n$ and < 0 for some other $\boldsymbol{x}$ in $\mathbb{R}^n$.*

Remark. A matrix that is positive (negative) definite is automatically positive (negative) semidefinite. Otherwise, every symmetric matrix falls into one of the above five categories.

For a simpler notation, let $a_{11} = a$ and $a_{12} = a_{21} = b$ and $a_{22} = c$. If $a = c = 0$ then the quadratic form is indefinite: $x_1x_2 > 0$ for some vectors $\mathbf{x}$ and $x_1x_2 < 0$ for other vectors $\mathbf{x}$. In fact, if just one of the coefficients a or c is zero, the quadratic form is indefinite: for example, if $a = 0$ then $x^TAx = 2bx_1x_2 + cx_2^2 = (2bx_1 + cx_2)x_2$, so the sign of x^TAx depends on the sign of $2bx_1 + cx_2$, which will clearly be positive for some vectors $\mathbf{x}$ and negative for others.

Let's assume that both $a \neq 0$ and $c \neq 0$.

$$\begin{aligned}
\mathbf{x}^TA\mathbf{x} &= ax_1^2 + 2bx_1x_2 + cx_2^2 \\
&= a(x_1^2 + 2\frac{b}{a}x_1x_2) + cx_2^2 + \frac{b^2}{a}x_2^2 - \frac{b^2}{a}x_2^2 \\
&= a(x_1^2 + 2\frac{b}{a}x_1x_2 + \frac{b^2}{a^2}x_2^2) + \big(c - \frac{b^2}{a^2}\big)x_2^2 \\
&= a\big(x_1 + \frac{b}{a}x_2\big)^2 + \frac{1}{a}(ac - b^2)x_2^2 \\
&= a\big(x_1 + \frac{b}{a}x_2\big)^2 + \frac{1}{a}|A|x_2^2.
\end{aligned}$$

It is clear that if $a > 0$ and $|A| > 0$, then $\mathbf{x}^TA\mathbf{x}$ is positive definite. Notice that although we assumed that a and c are both nonzero, we didn't actually use the fact that $c \neq 0$. However, one of the conditions for definiteness (positive or negative) is that $|A| > 0$ and this requires that $ac > 0$, i.e., that a and c have the same sign. Also, note that we could have carried out the above argument with the roles of a and c reversed.

Theorem 1.3. *A 2×2 symmetric matrix A is*

- *positive definite if and only if $|A| > 0$ and either $a_{11} > 0$ or $a_{22} > 0$, which is equivalent to $|A| > 0$ and both $a_{11} > 0$ and $a_{22} > 0$;*

- *negative definite if and only if $|A| > 0$ and either $a_{11} < 0$ or $a_{22} < 0$, which is equivalent to $|A| > 0$ and both $a_{11} < 0$ and $a_{22} < 0$.*

What are necessary and sufficient conditions for the matrix A and the associated quadratic form xAx to be positive or negative semidefinite? If A is positive semidefinite , i.e., $xAx \geq 0$ for all $x \neq 0$, we clearly must have $a \geq 0$ and $c \geq 0$. If either $a > 0$ or $c > 0$, then we must have $|A| \geq 0$; and if $a = c = 0$, then we must have $b = 0$ as well, in which case $|A| =$. Therefore, the conditions, $a \geq 0$, $c \geq 0$, and $|A| \geq 0$ together must all hold if A is positive semidefinite. Are these conditions also sufficient? Suppose that $|A| \geq 0$. If $a = c = 0$, then $|A| \geq 0$ implies that $b = 0$, so that A is the zero matrix, and $xAx = 0$ for all $x \in \mathbb{R}^2$. And it is clear in the above expression for xAx that if $a > 0$ (or symmetrically, $c > 0$) and also

$|A| \geq 0$, then $xAx \geq 0$ for all $x \neq 0$. Therefore, the conditions $a \geq 0$, $c \geq 0$ and $|A| \geq 0$ are sufficient as well as necessary for A to be positive semidefinite. A parallel argument provides the conditions for A to be negative semidefinite, and we have the following theorem.

Theorem 1.4. *A 2×2 symmetric matrix A is*

- *positive semidefinite if and only if $a_{11} \geq 0$, $a_{22} \geq 0$ and $|A| \geq 0$;*
- *negative semidefinite if and only if $a_{11} \leq 0$, $a_{22} \leq 0$, and $|A| \geq 0$.*

This pattern generalizes to $\mathbb{R}^n$ for arbitrary n as follows: first think of the components a_{ij} of the 2×2 matrix A as 1×1 "submatrices" of A; because they are 1×1 we call them "order-1" submatrices, and we will say that A itself, which is 2×2, is an order-2 submatrix. For an $n \times n$ matrix A we will say that a submatrix of order k consists of k of the rows and k of the columns of A, or we could equivalently say that an order-k submatrix is formed by deleting $n - k$ rows and columns. Now note that in the 2×2 example, the conditions we developed involved only submatrices on the diagonal a_{11} and a_{22} as well as A. We could say each of these submatrices was formed by deleting the same row and column: for a_{11} we deleted the second row and the second column; for a_{22} we deleted the first row and column; for A itself we deleted no rows and columns. Finally, note that the conditions in the 2×2 case were conditions on the signs of the determinants of these submatrices. The following definition generalizes these ideas to $n \times n$ matrices.

Definition 1.5. *Let A be an $n \times n$ matrix. For each $k = 1, 2, \cdots, n$ an **order-k principal submatrix** of A is a $k \times k$ matrix formed by deleting the same $n - k$ rows and columns. The order-k submatrices formed by deleting the last $n - k$ rows and columns are called the **leading** principal submatrices of A. The determinant of a submatrix of A is called a **minor** of A.*

Therefore the leading principal submatrices of a 2×2 matrix A are the matrices a_{11} and A itself; the leading principal minors are a_{11} and $|A|$. The leading principal minors of a 3×3 matrix A are

$$a_{11} \text{ and } \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \text{ and } |A|.$$

The leading principal minors of a 4×4 matrix A are

$$a_{11} \text{ and } \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \text{ and } \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \text{ and } |A|.$$

The general versions of our 2×2 theorems are as follows:

Theorem 1.6. *An $n \times n$ symmetric matrix is:*

- *positive definite if and only if all of its n leading principal minors are (strictly) positive;*
- *negative definite if and only if its n leading principal minors alternate in sign as follows:*

$$|A_1| < 0, \;\; |A_2| > 0, \;\; |A_3| < 0, \;\; \textit{etc.}$$

The k-th order leading principal minor should have the same sign as $(-1)^k$.

Theorem 1.7. *An $n \times n$ symmetric matrix is*

- *positive semidefinite if and only if all of its principal minors are non-negative;*
- *negative semidefinite if and only if all of every principal minor of odd order is non-positive and every principal minor of even order is non-negative.*

Corollary 1.7.1. *An $n \times n$ symmetric matrix is indefinite if some k-th order leading principal minor of A (or some pair of them) is nonzero but does not fit either of the above two sign patterns. (This case occurs when A has a negative k-th order leading principal minor for an even integer k or when A has a negative k-th order leading principal minor and a positive l-th order leading principal minor for two distinct odd integers k and l.)*

Example. In order that the matrix A be positive definite it is necessary and sufficient that the leading principal minors all be positive. If therefore seems natural to think that the parallel result should hold for positive *semidefiniteness*: that A is positive semidefinite if and only if all of its leading principal minors are nonnegative. Here is a counterexample, which shows that merely having nonnegative *leading* principal minors is not sufficient to ensure that A is positive semidefinite: we need to consider all the principal minors.

Let $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 2 \end{bmatrix}$. All three leading principal minors are either positive or zero:

$$|A_1| = a_{11} = 1, \;\; |A_2| = \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = 0, \;\text{and}\; |A_3| = |A| = 0.$$

However, the order-2 non-leading principal minor

$$\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 2 & 2 \end{vmatrix} = -2$$

which is inconsistent with both positive and negative semidefiniteness: it is negative, which is inconsistent with positive semidefiniteness; and its order is $k = 2$ and its sign is $-1 \neq (-1)^k$, which is inconsistent with negative semidefiniteness. Note that $\mathbf{x}^T A\mathbf{x} = x_1^2 + 2x_3^2 + 4x_1x_3$. When $\mathbf{x} = (1, 0, 1)$, then $\mathbf{x}^T A\mathbf{x} = 7$; when $\mathbf{x} = (1, 0, -1)$, then $\mathbf{x}^T A\mathbf{x} = -1$; this verifies directly, without having to consider principal minors, that A is neither negative semidefinite nor positive semidefinite - i.e., that it is indefinite.

Definiteness of a quadratic form is relevant to the optimization problem. This is because the definiteness of a quadratic form determines if $\mathbf{x} = \mathbf{0}$ is maxima or minima. In Economic problem, however, we encounter many constraints, which needs the optimization problem with constraints.

When the minimization/maximization problem of quadratic form involves linear constraints, the solution is slightly different. Check the below example to see how the definiteness of quadratic forms change when it is restricted to linear subspaces of $\mathbb{R}^n$:

Example.

$$Q(x_1, x_2) = ax_1^2 + 2bx_1x_2 + cx_2^2 = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

subject to the linear constraint,

$$Ax_1 + Bx_2 = 0.$$

If we set $x_1 = -Bx_2/A$ and replace x_1 with this value, the problem is simpler:

$$\begin{aligned} Q\Big(-\frac{Bx_2}{A}, x_2\Big) &= a\Big(-\frac{Bx_2}{A}\Big)^2 + 2b\Big(-\frac{Bx_2}{A}\Big)x_2 + cx_2^2 \\ &= \frac{aB^2}{A^2}x_2^2 - \frac{2bB}{A}x_2^2 + cx_2^2 \\ &= \frac{aB^2 - 2bAB + cA^2}{A^2}x_2^2. \end{aligned}$$

Then, Q is positive definite if and only if $aB^2 - 2bAB + cA^2 > 0$ and negative definite if and only if $aB^2 - 2bAB + cA^2 < 0$. However, this term can also be written as:

$$aB^2 - 2bAB + cA^2 = -\det \begin{pmatrix} 0 & A & B \\ A & a & b \\ B & b & c \end{pmatrix}.$$

Following the above example, we write below theorem.

Theorem 1.8. *The quadratic form $Q(x_1, x_2) = ax_1^2 + 2bx_1x_2 + cx_2^2$ is positive definite (negative definite) on the constraint set $Ax_1 + Bx_2 = 0$ if and only if*

$$\det \begin{pmatrix} 0 & A & B \\ A & a & b \\ B & b & c \end{pmatrix}$$

is negative (positive).

Generalizing this theorem to $\mathbb{R}^n$ linear subspace, we have:

Theorem 1.9. *To determine the definiteness of $Q(\boldsymbol{x}) = \boldsymbol{x}^T A \boldsymbol{x}$, when restricted to a constraint set given by m linear equations $B\boldsymbol{x} = \boldsymbol{0}$, construct the $(n+m) \times (n+m)$ symmetric matrix H by bordering the matrix A above and to the left by the coefficient B of the linear constraints:*

$$H = \begin{pmatrix} 0 & B \\ B^T & A \end{pmatrix}.$$

check the signs of the last $n-m$ leading principal minors of H, starting with the determinant of H itself.

(a) If $\det H$ has the same sign as $(-1)^n$ and if these last $n-m$ leading principal minors alternate in sign, then Q is negative definite on the constraint set $B\boldsymbol{x} = \boldsymbol{0}$, and $\boldsymbol{x} = \boldsymbol{0}$ is a strict global max of Q on this constraint set.

(b) If $\det H$ and these last $n-m$ leading principal minors all have the same sign as $(-1)^m$, then Q is positive definite on the constraint set $B\boldsymbol{x} = \boldsymbol{0}$, and $\boldsymbol{x} = \boldsymbol{0}$ is a strict global min of Q on this constraint set.

(c) If both of these conditions (a) and (b) are violated by nonzero leading principal minors, then Q is indefinite on the constraint set $B\boldsymbol{x} = \boldsymbol{0}$, and $\boldsymbol{x} = \boldsymbol{0}$ is neither a max nor a min of Q on this constraint set.

2 Unconstrained Optimization

Definition 2.1. *Let $f : X \to \mathbb{R}$ be a real-valued function and let $\bar{x} \in X$. We say that*

- *$\bar{x}$ is a (global) **maximum** or **maximum point** of f if $\forall x \in X$: $f(\bar{x}) \geq f(x)$; and*
- *$\bar{x}$ is a (global) **minimum** or **minimum point** of f if $\forall x \in X : f(\bar{x}) \leq f(x)$.*

*We say that $\bar{x}$ is an **extremum** if it is either a maximum or a minimum.*

Definition 2.2. *Let $f : X \to \mathbb{R}$ be a real-valued function on a set X in $\mathbb{R}^n$ and let $\bar{x} \in X$. We say that*

- *$\bar{x}$ is a **local maximum** of f if there is a neighborhood $\mathcal{N}$ of $\bar{x}$ such that $\forall x \in \mathcal{N}$: $f(\bar{x}) \geq f(x)$;*
- *$\bar{x}$ is a **local minimum** of f if there is a neighborhood $\mathcal{N}$ of $\bar{x}$ such that $\forall x \in \mathcal{N}$: $f(\bar{x}) \leq f(x)$.*

*We say that $\bar{x}$ is a **locally unique** or **strict** maximum or minimum if there is a neighborhood $\mathcal{N}$ in which the inequality is strict for all $x \in \mathcal{N}$ except $\bar{x}$ itself.*

Obviously, a global maximum/minimum is a local maximum/minimum, and a local maximum/minimum is not necessarily a global maximum/minimum.

Definition 2.3. *A point $\bar{\boldsymbol{x}} \in \mathbb{R}^n$ is a **critical point** of a function $f : \mathbb{R}^n \to \mathbb{R}$ if $\nabla f(\bar{\boldsymbol{x}}) = \boldsymbol{0}$, i.e., if all partial derivatives of f at $\bar{x}$ are zero.*

Our first theorem establishes a necessary condition for a point to be a maximum or a minimum of a function: the point must be a critical point. We first state and prove the theorem for real functions, and only for a local maximum. The proof is easily altered to cover the alternative case of a local minimum. Then we state and prove the theorem for functions on $\mathbb{R}^n$.

Theorem 2.4. *Let $f : \mathbb{R} \to \mathbb{R}$. If $\bar{x}$ is a local maximum of f and f is differentiable at $\bar{x}$, then $\bar{x}$ is a critical point of f.*

Proof. Suppose that $f'(\bar{x}) > 0$. Then there is a $\delta > 0$ such that

$$0 < |\Delta x| < \delta \Rightarrow \frac{f(\bar{x} + \Delta x) - f(\bar{x})}{\Delta x} > 0$$

Therefore, for $0 < \Delta x < \delta$, we have $f(\bar{x} + \Delta x) - f(\bar{x}) > 0$, i.e., $f(\bar{x} + \Delta x) > f(\bar{x})$. In every neighborhood of $\bar{x}$ there are points $\bar{x} + \Delta x) > f(\bar{x})$, so $\bar{x}$ is not a local maximum of f, contrary to the theorem's assumption. Therefore, $f'(\bar{x}) > 0$ cannot be true. A parallel argument shows that $f'(\bar{x}) < 0$ also cannot be true. Since $f'(\bar{x})$ does exist, we have $f'(\bar{x}) = 0$. $\square$

While a local maximum or minimum point must be a critical point - a necessary condition - the reverse is not ture: "$\bar{x}$ is a critical point" is not sufficient to ensure that $\bar{x}$ is either a maximum or minimum point. The classical counterexample is $f(x) = x^3$: $\bar{x} = 0$ is a critical point of f, but is obviously neither a local maximum nor a local minimum.

We often refer to properties of a function's derivative as "first-order" properties, and properties of a function's second derivative as "second-order" properties of the functions. So here we say that a first-order necessary condition for $\bar{x}$ to be a local maximum or minimum of a function is that it is a critical point of the function.

Now we obtain the generalization to multivariate functions, the first-order necessary condition for a local maximum of a function $f : \mathbb{R}^n \to \mathbb{R}$. As with $n = 1$ case above, the proof is easily altered to cover the case of a local minimum.

Theorem 2.5. *Let $f : \mathbb{R}^n \to \mathbb{R}$. If $\bar{\boldsymbol{x}} \in \mathbb{R}^n$ is a local maximum of f and f is differentiable at $\bar{\boldsymbol{x}}$, then $\bar{\boldsymbol{x}}$ is a critical point of f.*

Proof. We apply the same argument as in the preceding proof to each of the partial derivatives of f. Suppose $\partial f(\bar{\mathbf{x}})/\partial x_i > 0$. Then there exists a $\delta > 0$ such that

$$0 < |h| < \delta \Rightarrow \frac{f(\bar{\mathbf{x}} + h\mathbf{e}_i) - f(\bar{\mathbf{x}})}{h} > 0,$$

where $\mathbf{e}_i$ is the i-th unit vector in $\mathbb{R}^n$. Then for $0 < h < \delta$ we have $f(\bar{\mathbf{x}} + h\mathbf{e}_i) - f(\bar{\mathbf{x}}) > 0$, i.e., $f(\bar{\mathbf{x}} + h\mathbf{e}_i) > f(\bar{\mathbf{x}})$. Therefore there is no neighborhood of $\bar{\mathbf{x}}$ on which $f(\mathbf{x}) \leq f(\bar{\mathbf{x}})$, so $\bar{\mathbf{x}}$ is not a local maximum of f. Since this contradicts the assumption of the theorem, we cannot have $\partial f(\bar{\mathbf{x}})/\partial x_i > 0$. A parallel argument shows that we cannot have $\partial f(\bar{\mathbf{x}})/\partial x_i < 0$. Therefore, $\partial f(\bar{\mathbf{x}})/\partial x_i = 0$. □

Now we develop a sufficient condition for a local extremum, which includes a second-order condition as well as the first-order critical-point condition - a condition on the function's second derivative. We again begin with the case $n = 1$ before dealing with the case of an arbitrary n. And this time we prove the theorem for the case of a local minimum; the proof for a local maximum is a straightforward alteration of this proof.

Theorem 2.6. *Let $f : \mathbb{R} \to \mathbb{R}$ and let $\bar{x}$ be a point at which f is twice differentiable. If $f'(\bar{x}) = 0$ and $f''(\bar{x}) > 0$, then $\bar{x}$ is a local minimum point of f.*

Proof. Define the function F by $F(\Delta x) = f(\bar{x} + \Delta x) - f(\bar{x})$. We will show that for some neighborhood $\mathcal{N}$ around zero

$$[\Delta x \in \mathcal{N} \& \Delta x \neq 0] \Rightarrow F(\Delta x) > 0,$$

i.e., $f(\bar{x} + \Delta x) > f(\bar{x})$ for all nonzero $\Delta x \in \mathcal{N}$, so this will show that $\bar{x}$ is indeed a locally unique minimum point of f.

Using the 2nd degree Taylor polynomial, we have

$$F(\Delta x) = f'(\bar{x})\Delta x + \frac{1}{2} f''(\bar{x})(\Delta x)^2 + R_2(\Delta x) \tag{2}$$

where $R_2(\Delta x)$ is the remainder term, which satisfies $\lim_{\Delta x \to 0} \frac{1}{(\Delta x)^2} R_2(\Delta x) = 0$. Because the theorem assumes that $f'(\bar{x}) = 0$, (2) becomes

$$F(\Delta x) = \frac{1}{2} f''(\bar{x})(\Delta x)^2 + R_2(\Delta x) \tag{3}$$

and because we are excluding $\Delta x = 0$, we can divide both sides of equation (3) by $(\Delta x)^2$, so we have

$$\frac{1}{(\Delta x)^2} F(\Delta x) = \frac{1}{2} f''(\bar{x}) + \frac{1}{(\Delta x)^2} R_2(\Delta x). \tag{4}$$

(4) is the key element of the proof: the two terms on the right-hand side are $\frac{1}{2} f''(\bar{x})$, which is a given positive number, and $\frac{1}{(\Delta x)^2} R_2(\Delta x)$, which is very small when Δx is small. So for small values of Δx, the first term dominates the second term - i.e., since the first term is a positive number, when we add the second term to it (even if the second term is negative), the RHS will still be positive. (Note, in fact, that when $R_2(\Delta x)$ is positive, the RHS is clearly positive, so it is only those Δx for which $R_2(\Delta x)$ is negative that we have to worry about.) This will guarantee that $F(\Delta x)$ on the LHS is positive, which is the conclusion we are trying to establish.

Now let's write down formally what we described informally in the preceding paragraph. Since $\lim_{\Delta x \to 0} \frac{1}{(\Delta x)^2} R_2(\Delta x) = 0$, there is a $\delta > 0$ such that

$$0 < |\Delta x| < \delta \Rightarrow \left| \frac{1}{(\Delta x)^2} R_2(\Delta x) \right| < \frac{1}{2} f''(\bar{x})$$

Therefore, when $0 < |\Delta x| < \delta$, we have

$$\frac{1}{2} f''(\bar{x}) + \frac{1}{(\Delta x)^2} R_2(\Delta x) > 0,$$

and therefore $F(\Delta x) > 0$, i.e., $f(\bar{x} + \Delta x) > f(\bar{x})$, for all Δx such that $0 < |\Delta x| < \delta$, completing the proof.

□

Now we want to prove the theorem for functions on any Euclidean space $\mathbb{R}^n$. Conceptually, we can copy the $n = 1$ proof we just completed. The key idea of the proof will still be the idea expressed in equation (4).

But because the displacement Δx will be vectors rather than just numbers, three features of the proof will have to be adjusted: i we can't divide by $(\Delta x)^2$ when Δx is a vector; ii when we work with the limit of a function on $\mathbb{R}^n$, we are not just working with numbers that are small, but with vectors that are small, i.e., whose lengths are small; and iii the second-derivative term in equation (4) - the first term on the RHS - won't be a number, but the Hessian matrix of f at $\bar{x}$ and we will have to turn that into a number for the equation to make sense.

Theorem 2.7. *Let $f : \mathbb{R}^n \to \mathbb{R}$ and let $\bar{\boldsymbol{x}}$ be a point at which f is twice differentiable. If $\nabla f(\bar{\boldsymbol{x}}) = \boldsymbol{0}$ and the Hessian matrix $H_f(\bar{\boldsymbol{x}})$ is positive definite, then $\bar{\boldsymbol{x}}$ is a locally unique minimum point of f.*

We omit the proof for the sufficient condition in $\mathbb{R}^n$ since it is similar to the sufficient condition in $\mathbb{R}$. However, we need to use the Weierstrass Theorem, which guarantees a local maximum and a local minimum of a continuous function that is defined on a closed and bounded subset of $\mathbb{R}^n$. Similar to Theorem 2.7, we have a parallel theorem that provides a sufficient condition for a point $\bar{x}$ to be a local minimum.

Theorem 2.8. *Let $f : \mathbb{R}^n \to \mathbb{R}$ and let $\bar{\boldsymbol{x}}$ be a point at which f is twice differentiable. If $\nabla f(\bar{\boldsymbol{x}}) = \boldsymbol{0}$ and the Hessian matrix $H_f(\bar{\boldsymbol{x}})$ is negative definite, then $\bar{\boldsymbol{x}}$ is a locally unique maximum point of f.*

This follows immediately from the local-minimum theorem: $\bar{x}$ is a local maximum of f if and only if it is a local minimum of $-f$; and the Hessian of $-f$ is the negative definite of the Hessian of f, so the associated quadratic forms satisfy $\Delta\mathbf{x}^T H_{-f}(\bar{\mathbf{x}})\Delta\mathbf{x} = -\Delta\mathbf{x}^T H_f(\bar{\mathbf{x}})\Delta\mathbf{x}$.

We have developed both a sufficient condition to ensure that a point $\bar{x}$ is a local maximum or minimum (consisting of a first-order condition together with a second-order condition), and a necessary first-order condition that a point has to satisfy if it is a local maximum or minimum. But it seems clear that there is also a second-order condition that a local maximum $\bar{x}$ has to satisfy. The condition cannot be as strong as the second-order sufficient condition. The second-order necessary condition for a point to be a local maximum or minimum is the slightly weaker condition that the Hessian matrix must be negative or positive semidefinite instead of definite:

Theorem 2.9. *Let $f : \mathbb{R}^n \to \mathbb{R}$ and let $\bar{\boldsymbol{x}}$ be a point at which f is twice differentiable. If $\bar{x}$ is a local maximum of f, then $\nabla f(\bar{\boldsymbol{x}}) = \boldsymbol{0}$ and the Hessian matrix $H_f(\bar{\boldsymbol{x}})$ is negative semidefinite. If $\bar{x}$ is a local minimum of f, then $\nabla f(\bar{\boldsymbol{x}}) = \boldsymbol{0}$ and the Hessian matrix $H_f(\bar{\boldsymbol{x}})$ is positive semidefinite.*

So far we have only talked about the conditions for a local maximum and a local minimum. However, local minimum/maximum does not guarantee that it is global minimum/maximum. The conditions for a global minimum/maximum is related to if the function is concave or convex function. Indeed if function $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ is concave (convex) function on a subset U in $\mathbb{R}^n$ and $Df(\mathbf{x}^*) = \mathbf{0}$ for some $\mathbf{x}^* \in U$, then $\mathbf{x}^*$ is a **global max (global min)** of f on U. We discuss more in detail in the section of concave/convex function.

3 Constrained Optimization

In Economic problems, we mostly encounter the problem with constraints. For example, the maximization problem of firm's profit can be constrained by the availability of inputs and the cost. The problem can have constraints either equality constraints or inequality constraints, or sometimes both of these constraints. Generally, we write the problem as:

$$\begin{aligned} \max & f(x_1, \cdots, x_n) \text{ s.t.} \\ & g_1(x_1, \cdots, x_n) \leq b_1, \cdots, g_k(x_1, \cdots, x_n) \leq b_k \\ & h_1(x_1, \cdots, x_n) = c_1, \cdots, h_m(x_1, \cdots, x_n) = c_m. \end{aligned}$$

In this problem, we call f as **objective function**, the g_j's **inequality constraints** and the h_i's **equality constraints**. In Economic problems, we have **nonnegativity constraints**, such as $x_i \geq 0$ for $i = 1, \cdots, n$.

3.1 Equality constraints

For example, let us have the maximization problem of two inputs, such as utility maximization problem.

$$\max f(x_1, x_2) \text{ s.t. } p_1 x_1 + p_2 x_2 = I$$

What we are interested in is to find $x^* = (x_1^*, x_2^*)$ that maximizes our function f that satisfies the binding budget set, $p_1 x_1 + p_2 x_2 = I$. Geometrically, this point can be found as the point where the slope of f at x^* is same as the slope of the constraint set at x^*. Formulating the geometrical view as equations gives us the **Lagrangian function** as:

$$L(x_1, x_2, \mu) \equiv f(x_1, x_2) - \mu(h(x_1, x_2) - c).$$

where μ is Lagrangian multiplier and $h(x_1, x_2) = c$ is our constraint set. By having Lagrangian function, we transform the problem with constraint into the problem of three variables without constraint. Generalizing this idea to a problem of n variables with m equality constraints, the Lagrangian function solves the problem as below theorem:

Theorem 3.1. *Let $f, h_1, \cdots, h_m$ be C^1 function of n variables. Consider the problem of maximizing (or minimizing) $f(\boldsymbol{x})$ on the constraint set*

$$C_h \equiv \{\boldsymbol{x} = (x_1, \cdots, x_n) : h_1(\boldsymbol{x}) = a_1, \cdots, h_m(\boldsymbol{x}) = a_m\}.$$

Suppose that $\boldsymbol{x}^* \in C_h$ and that $\boldsymbol{x}^*$ is a (local) max or min of f on C_h. Suppose further that $\boldsymbol{x}^*$ satisfies the rank of $Dh(\boldsymbol{x}^*)$ be m. Then, there exists $\mu_1^*, \cdots, \mu_m^*$ such that $(x_1^*, \cdots, x_n^*, \mu_1^*, \cdots, \mu_m^*) \equiv (\boldsymbol{x}^*, \mu^*)$ is a critical point of the Lagrangian

$$L(\boldsymbol{x}, \mu) \equiv f(\boldsymbol{x}) - \mu_1[h_1(\boldsymbol{x}) - a_1] - \mu_2[h_2(\boldsymbol{x}) - a_2] - \cdots - \mu_m[h_m(\boldsymbol{x}) - a_m].$$

In other words,

$$\begin{aligned} &\frac{\partial L}{\partial x_1}(\boldsymbol{x}^*, \mu^*) = 0, \cdots, \frac{\partial L}{\partial x_n}(\boldsymbol{x}^*, \mu^*) = 0, \\ &\frac{\partial L}{\partial \mu_1}(\boldsymbol{x}^*, \mu^*) = 0, \cdots, \frac{\partial L}{\partial \mu_m}(\boldsymbol{x}^*, \mu^*) = 0. \end{aligned}$$

3.2 Inequality Constraints

When the constraints are defined by inequalities, finding a solution for maximization or minimization problem becomes more complicated. Now, we need to consider that the constraint might not be binding. Because of this, the Lagrangian multiplier needs to satisfy $\lambda[g(x, y) - b] = 0$, where $g(x, y) - b$ is the inequality constraint as $g(x, y) \leq b$. We call this condition as **complementary slackness condition**. To satisfy this condition, we have either $\lambda = 0$ or $g(x, y) = b$. (Geometrical explanation can be found in SB textbook, page 424 - 427.) Including this condition, we have the following theorem:

Theorem 3.2. *Suppose that f and g are C^1 functions on $\mathbb{R}^2$ and that (x^*, y^*) maximizes f on the constraint set $g(x, y) \leq b$. If $g(x^*, y^*) = b$, suppose that*

$$\frac{\partial g}{\partial x}(x^*, y^*) \neq 0 \text{ or } \frac{\partial g}{\partial y}(x^*, y^*) \neq 0.$$

In any case, form the Lagrangian function

$$L(x, y, \lambda) = f(x, y) - \lambda[g(x, y) - b].$$

Then, there is a multiplier λ^* *such that:*

(a) $\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*) = 0$,

(b) $\frac{\partial L}{\partial y}(x^*, y^*, \lambda^*) = 0$,

(c) $\lambda^*[g(x^*, y^*) - b] = 0$,

(d) $\lambda^* \geq 0$,

(e) $g(x^*, y^*) \leq b$.

Following this theorem, if $\lambda^* > 0$, then the inequality constraint will be binding and finding a solution becomes easier as it is same as in problem of equality constraints. The theorem also can be generalized to a function that is defined on $\mathbb{R}^n$ with k different inequality constraints.

Theorem 3.3. *Suppose that* $f, g_1, \cdots, g_k$ *are* C^1 *functions of* n *variables. Suppose that* $\boldsymbol{x}^* \in \mathbb{R}^n$ *is a local maximizer of* f *on the constraint set defined by the* k *inequalities,*

$$g_1(x_1, \cdots, x_n) \leq b_1, \cdots, g_k(x_1, \cdots, x_n) \leq b_k.$$

Then, we can form the Lagrangian as:

$$L(x_1, \cdots, x_n, \lambda_1, \cdots, \lambda_k) = f(\boldsymbol{x}) - \lambda_1[g_1(\boldsymbol{x}) - b_1] - \cdots - \lambda_k[g_k(\boldsymbol{x}) - b_k].$$

Then, there exist multipliers $\lambda_1^*, \cdots, \lambda_k^*$ *such that:*

(a) $\frac{\partial L}{\partial x_1}(\boldsymbol{x}^*, \lambda^*) = 0, \cdots, \frac{\partial L}{\partial x_n}(\boldsymbol{x}^*, \lambda^*) = 0$,

(b) $\lambda_1^*[g_1(\boldsymbol{x}^*) - b_1] = 0, \cdots, \lambda_k^*[g_k(\boldsymbol{x}^*) - b_k] = 0$,

(c) $\lambda_1^* \geq 0, \cdots, \lambda_k^* \geq 0$,

(d) $g_1(\boldsymbol{x}^*) \leq b_1, \cdots, g_k(\boldsymbol{x}^*) \leq b_k$.

In this theorem, if we have some binding constraints, say k_0 constraints are binding, then we need an assumption that the rank at $\mathbf{x}^*$ of the Jacobian matrix of the binding constraints is k_0.

λ_i for $i = 1, \cdots, m$ is the "shadow value" of the i-th constraint, the marginal value to the objective function of relaxing or tightening the constraint by one unit. For example, think $f(\mathbf{x})$ as the dollar value, e.g., the profit, of operating at the levels described by the vector $\mathbf{x}$ and think of the function $h_i(\mathbf{x})$ as the amount of some resource R_i required to operate at the vector $\mathbf{x}$ and the number a_i as the amount of the resource R_i that is available to the decision-maker. Then, λ_i is the marginal increase in profit $f(\mathbf{x})$ that will result from having one additional unit of resource R_i, the marginal value of the resource. If constraint i is non-binding, then the marginal value of an increase or decrease in a_i is zero, i.e., $\lambda_i = 0$. If constraint i is binding, then λ_i tells us how much $f(\mathbf{x})$ will be increased or decreased by a unit increase or decrease in availability of resource R_i.

3.3 Nonnegativity constraints

When we have nonnegativity constraints, we can solve the problem using Kuhn-Tucker (KT) conditions , based on the work of Harold Kuhn and A. W. Tucker. The Kuhn-Tucker conditions are first-order conditions for constrained optimization problems, a generalization of the first-order conditions we are already familiar with. These more general conditions provide a unified treatment of constrained optimization, in which we allow for inequality constraints, non-negativity constraints, and boundary solutions (some x_i's$= 0$) are permitted, etc. A special case covered by the Kuhn-Tucker conditions is linear programming.

Let $f : \mathbb{R}^n \to \mathbb{R}$ and $G : \mathbb{R}^n \to \mathbb{R}^m$ be continuously differentiable functions, and let $b \in \mathbb{R}^m$. We want to characterize those vectors $\mathbf{x}^* \in \mathbb{R}^n$ that is a solution of the problem as following:

$$\max f(\mathbf{x}) \text{ s.t. } \mathbf{x} \geq \mathbf{0} \text{ and } G(\mathbf{x}) \leq \mathbf{b}$$

The Kuhn-Tucker conditions are the first-order conditions that characterize the vectors $\mathbf{x}^*$ that is a solution of the problem: $\exists \lambda_1, \cdots, \lambda_m \in \mathbb{R}_+$ such that

(KT1) For $j = 1, \cdots, n$: $\frac{\partial f}{\partial x_j} \leq \sum_{i=1}^m \lambda_i \frac{\partial G^i}{\partial x_j}$, with equality if$x_j^* > 0$;

(KT2) For $i = 1, \cdots, m$: $G^*(\mathbf{x}^*) \leq b_i$, with equality if $\lambda_i > 0$,

An equivalent statement of the conditions: $\exists \lambda \in \mathbb{R}_+^m$ such that

(KT1) $\nabla f \leq \sum_{i=1}^m \lambda_i \nabla G^i$ and $\mathbf{x}^* \cdot (\nabla f - \sum_{i=1}^m \lambda_i \nabla G^i) = 0$;

(KT2) $G(\mathbf{x}^*) \leq \mathbf{b}$ and $\lambda \cdot (\mathbf{b} - G(\mathbf{x}^*)) = 0$.

where in both statements of conditions, the partial derivative and gradient are evaluated at $\mathbf{x}^*$.

4 The Envelope Theorem

In an optimization problem, we often want to know how the value of the objective function will change if one or more of the parameter values changes. Let's consider a simple example: a price-taking firm choosing the quantities of a single output that maximize their profit. If the market price of its product changes, how much will the firm's profit increase or decrease? At first, the answer seems obvious: if the price change is $\Delta p > 0$, an increase, then profit will increase by Δp on each unit the firm is producing, i.e., by $(\Delta p)q$ if the firm is producing q units.

But on second thought, this doesn't seem quite right. When the price increases, the firm will probably change the amount q that it produces, so it seems as if we can't just apply the price change to the q units it was producing, but we apparently need to also take account of the change in q as well. Check below example:

The firm's problem is:

$$\max_{q \in \mathbb{R}_+} \pi(q; p) = pq - C(q),$$

where $C(q)$ is the firm's cost function, which we will assume it is strictly convex and differentiable. For the firm, this is a problem with just one decision variable, q and one parameter, p. The first-order condition for (P_1) is

$$\frac{\partial \pi(q)}{\partial q} = 0 \text{ i.e., } C'(q) = p$$

the firm chooses the output level q at which its marginal cost is equal to the market price (which the firm takes as given: we assumed it's a price-taking firm).

Let's write the solution function as $\hat{q}(p)$: then the value function[1] is

$$v(p) = \pi(\hat{q}(p), p) = p\hat{q}(p) - C(\hat{q}(p)),$$

and we have

$$v'(p) = \frac{\partial \pi}{\partial p} + \frac{\partial \pi}{\partial q}\frac{d\hat{q}}{dp} = q + \frac{\partial \pi}{\partial q}\frac{\partial \hat{q}}{\partial p}$$

which does take account of the firm's output response to a price change, $\partial \hat{q}/\partial p$. But this response is multiplied by the term $\partial \pi/\partial q$, the change in profit that results from the change

[1] The value function give the value of the objective function as a function of the parameters. If our objective function is $f(x; \theta)$ and the solution is $x(\theta)$, then the value function is $v(\theta) = f(x(\theta), \theta)$.

in q. And at the optimal q that is zero; the first-order condition is $\partial\pi/\partial q = 0$. (To put it another way, since MC =p, an increase in q will increase cost and revenue by approximately the same amount, leaving profit virtually unchanged.) So it turns out that we do have $v'(p) = q$ after all, just as we first conjectured.

Theorem 4.1 (Envelope Theorem)**.** *For the maximization problem,* $\max_{x\in\mathbb{R}} f(x;\theta)$*, if* $f : \mathbb{R}^2 \to \mathbb{R}$ *is a* C^1 *function and if the solution function for the maximization problem is differentiable at* $\bar{\theta}$*, then the derivative of the value function satisfies* $v'(\theta) = \partial f/\partial\theta$ *at* $\bar{\theta}$*.*

Corollary 4.1.1. *If the solution function is differentiable on an open subset* $U \subseteq \mathbb{R}$*, then* $v'(\theta) = \partial f/\partial\theta$ *on* U*.*

Example. Let's consider the following maximization problem:

$$\begin{aligned}\max_{x\in\mathbb{R}} f(x,\theta) &= \theta - (x-\theta)^2,\ \ \theta \in \mathbb{R}\\ &= \theta - x^2 + 2\theta x - \theta^2\\ &= 2\theta x - x^2 + \theta - \theta^2\end{aligned}$$

Suppose θ is the amount of capital a firm is using, and x is the firm's level of output. Assume that $f(x,\theta)$ is the firm's profit when it produces x units using θ units of capital. If the level of capital is taken as given (a parameter) by the firm - for example, if the firm is making a short-run output decision and it is unable to vary its capital in the short run - then the solution $\hat{x}(\cdot)$ for this problem tells us the profit-maximizing output level, $\hat{x}(\theta)$, for the given amount of capital, θ; and the value function, $v(\cdot)$ gives us the maximum profit that can be achieved, $v(\theta)$, with the amount θ of capital. In other words, $\hat{x}(\theta)$ and $v(\theta)$ are the short-run optimal output and the short-run maximum profit.

How much can the firm increases its profit when, in the longer run, it can vary θ? The Envelope Theorem will tell us the answer. First note that $\partial f/\partial\theta = 2x + 1 - 2\theta$. Therefore, according to Envelope Theorem, we have $v'(\theta) = \frac{\partial f}{\partial\theta} = 2x + 1 - 2\theta$. This expression depends on x as well as on θ; in order to determine $v'(\theta)$ from θ alone, we need to determine how x depends on θ - we need to determine the solution function. The first-order condition for the maximization problem is

$$\frac{\partial f}{\partial x} = 2\theta - 2x = 2(\theta - x) = 0 \text{ at } x = \theta$$

Therefore, the solution function is $\hat{x}(\theta) = \theta$ and $v'(\theta) = 2\theta + 1 - 2\theta = 1$.

For this problem, of course, it is simple to obtain the value function directly:

$$v(\theta) = f(\hat{x}(\theta), \theta) = f(\theta, \theta) = \theta - (\theta - \theta)^2 = \theta.$$

Thus, again, $v'(\theta) = 1$.

Below theorem shows the Envelope Theorem for n variables and m parameters.

Theorem 4.2. *Assume that $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is a C^1 function and consider the maximization problem $\max_{x \in \mathbb{R}^n} f(\boldsymbol{x}; \theta)$. If the solution function, $\hat{\boldsymbol{x}} : \mathbb{R}^m \to \mathbb{R}^n$ is differentiable on an open set $U \subseteq \mathbb{R}^m$, then the partial derivatives of the value function satisfy*

$$\frac{\partial v}{\partial \theta_i} = \frac{\partial f}{\partial \theta_i} \text{ on } U, \ \ i = 1, \cdots, m$$

Proof. At any $\theta \in U$, for each $i = 1, \cdots, m$, we have

$$\begin{aligned}\frac{\partial v}{\partial \theta_i} &= \frac{\partial}{\partial \theta_i} f(\hat{\mathbf{x}}(\theta), \theta) \\ &= \frac{\partial f}{\partial \theta_i} + \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} \frac{\partial \hat{x}_j}{\partial \theta_i} \\ &= \frac{\partial f}{\partial \theta_i}, \ \left(\because \frac{\partial f}{\partial x_j} = 0 \text{ at } \hat{\mathbf{x}}(\theta), \ j = 1, \cdots, n, \text{ by F.O.C.} \right)\end{aligned}$$

□

Now let's add a constraint to the maximization problem:

$$\max_{x \in \mathbb{R}^n} f(\mathbf{x}; \theta) \text{ subject to } G(\mathbf{x}; \theta) \leq 0$$

where $f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$ and $G : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$ are C^1-functions on an open set $U \subseteq \mathbb{R}^n \times \mathbb{R}$, and we assume that the solution function $\hat{\mathbf{x}}(\theta)$ exists and is differentiable for all $(\mathbf{x}, \theta) \in U$.

The first-order condition is, for some $\lambda \geq 0$,

$$\nabla f(\mathbf{x}; \theta) = \lambda \nabla G(\mathbf{x}; \theta) \text{ i.e., } \frac{\partial f}{\partial x_j}(\mathbf{x}; \theta) = \lambda \frac{\partial G}{\partial x_j}(\mathbf{x}; \theta), \ j = 1, \cdots, n.$$

Let's assume that $\lambda > 0$; otherwise either the constraint is not binding or $\nabla f(\mathbf{x}; \theta) = \mathbf{0}$, so we are back in the unconstrained case.

Because the constraint is binding, the solution function $\hat{\mathbf{x}}(\cdot)$ satisfies $G(\hat{\mathbf{x}}(\theta), \theta) = 0$ for all θ such that $((\hat{\mathbf{x}}(\theta), \theta) \in U$. We therefore have the following proposition, which will be instrumental in the proof of the Envelope Theorem.

Proposition 4.3. *If $G : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$ is a C^1 function and $\hat{\boldsymbol{x}}(\cdot) : \mathbb{R} \times \mathbb{R}^n$ satisfies $G(\hat{\boldsymbol{x}}(\theta), \theta) = 0$ for all $(x, \theta) \in U$, then*

$$\sum_{j=1}^{n} \frac{\partial G}{\partial x_j} \frac{\partial \hat{x}_j}{\partial \theta} = -\frac{\partial G}{\partial \theta}.$$

Proof. The 1st-degree Taylor polynomial equation for $\Delta G(\Delta \mathbf{x}, \Delta\theta) = 0$ is,

$$\frac{\partial G}{\partial x_1} \Delta x_1 + \cdots + \frac{\partial G}{\partial x_n} \Delta x_n + R(\Delta x, \Delta\theta) = 0,$$

where $\lim_{(\Delta x, \Delta\theta)} \frac{1}{\|(\Delta x, \Delta\theta)\|} R(\Delta \mathbf{x}, \Delta\theta) = 0$. Thus,

$$\begin{aligned} & \lim_{\Delta\theta \to 0} \Big[\sum_{j=1}^{n} \frac{\partial G}{\partial x_j} \frac{\Delta x_j}{\Delta \theta} + \frac{\partial G}{\partial \theta} \Big] = 0 \\ \Leftrightarrow \quad & \sum_{j=1}^{n} \frac{\partial G}{\partial x_j} \lim_{\Delta\theta \to 0} \frac{\Delta x_j}{\Delta \theta} = -\frac{\partial G}{\partial \theta} \\ \Leftrightarrow \quad & \sum_{j=1}^{n} \frac{\partial G}{\partial x_j} \frac{\partial \hat{x}_j}{\partial \theta} = -\frac{\partial G}{\partial \theta} \end{aligned}$$

□

Theorem 4.4 (Envelope Theorem)**.** *Assume that $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ and $G : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ are C^1-functions, and consider the maximization problem:*

$$\max_{\boldsymbol{x} \in \mathbb{R}^n} f(\boldsymbol{x}; \theta) \ \ s.t. \ G(\boldsymbol{x}, \theta) \leq 0.$$

Assume that the solution function $\hat{\boldsymbol{x}}(\cdot) : \mathbb{R}^m \to \mathbb{R}^n$ is differentiable at $\bar{\theta} \in \mathbb{R}^m$ and that λ is the value of the Lagrangian multiplier in the first-order condition for the problem at $(\hat{x}(\bar{\theta}), \bar{\theta})$. Then the partial derivatives of the value function satisfy,

$$\frac{\partial v}{\partial \theta_i} = \frac{\partial f}{\partial \theta_i} - \lambda \frac{\partial G}{\partial \theta_i} \ \ at \ \bar{\theta}, \ \ i = 1, \cdots, m,$$

where the partial derivatives of f and G are evaluated at $(\hat{\boldsymbol{x}}(\bar{\theta}), \bar{\theta})$.

Proof. For each $i = 1, \cdots, m$ and evaluating all the partial derivatives of f and G at $(\hat{x}(\bar{\theta}), \bar{\theta})$, we have

$$\begin{aligned}
\frac{\partial v}{\partial \theta_i}(\bar{\theta}) &= \frac{\partial f}{\partial \theta_i} + \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}\frac{\partial \hat{x}_j}{\partial \theta}(\bar{\theta}) \\
&= \frac{\partial f}{\partial \theta_i} + \sum_{j=1}^{n} \lambda \frac{\partial G}{\partial x_j}\frac{\partial \hat{x}_j}{\theta}(\bar{\theta}) \\
&= \frac{\partial f}{\partial \theta_i} - \lambda \frac{\partial G}{\partial \theta_i}
\end{aligned}$$

□

Theorem 4.5 (Generalized Envelope Theorem)**.** *Assume that* $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ *and* $G : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^l$ *are* C^1*-functions, and consider the maximization problem*

$$\max_{x \in \mathbb{R}^n} f(\boldsymbol{x}; \theta) \text{ s.t. } G(\boldsymbol{x}; \theta) \leq \boldsymbol{0}$$

where $G(\boldsymbol{x}; \theta) \leq \boldsymbol{0}$ *is equivalent to having* $G_1(\boldsymbol{x}, \theta) \leq 0, \cdots, G_l(\boldsymbol{x}, \theta) \leq 0$*. Assume that the solution function* $\hat{\boldsymbol{x}}(\cdot) : \mathbb{R}^m \to \mathbb{R}^n$ *is differentiable at* $\bar{\theta} \in \mathbb{R}^m$ *and that for each* $k = 1, \cdots, l$*,* λ_k *is the value of the Lagrangian multiplier for constraint* $G_k(\boldsymbol{x}, \theta) \leq 0$ *in the first-order condition at* $(\hat{\boldsymbol{x}}(\bar{\theta}), \bar{\theta})$*. Then, the partial derivatives of the value function satisfy:*

$$\frac{\partial v}{\partial \theta_i} = \frac{\partial f}{\partial \theta_i} - \sum_{k=1}^{l} \lambda_k \frac{\partial G_k}{\partial \theta_i} \text{ at } \bar{\theta},\ i = 1, \cdots, m,$$

where the partial derivatives of f *and* G *are evaluated at* $(\hat{\boldsymbol{x}}(\bar{\theta}), \bar{\theta})$*.*